

Robert Lee

Senior Software Engineer — Search, Systems & AI

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SUMMARY

Senior Software Engineer on Microsoft Azure AI Search building retrieval systems that ground enterprise AI in governed knowledge at billion-vector scale. Experience spans vector search, agentic retrieval, information retrieval, RAG, distributed systems, graph databases, performance optimization, and production reliability.

TECHNICAL SKILLS

Languages: C++, C#, Java, Python, TypeScript

Search & AI: Vector search, vector quantization, information retrieval, hybrid search, agentic retrieval, RAG, LLM agents, graph search, computer vision

Systems: Distributed systems, durable systems, performance engineering, SIMD, authentication, benchmarking, production diagnostics

EXPERIENCE

Senior Software Engineer

Mar. 2025 – Present

Microsoft — Azure AI Search C++, C#, Java, Python

Redmond, WA

- Engineering lead for a **vector-search diversity capability**, owning scoping, design, and implementation. Adapted Microsoft Research work to broaden evidence for complex queries at **billions-of-vectors scale**.
- Led **agentic filter and boost generation** from research to production. Reframed unbounded synthesis as a bounded, verifiable operator set; validated coverage against customer usage, aligned research and product teams, and shipped improvements to Public Preview.
- Integrated retrieval into tool-calling and multi-agent workflows, grounding LLM agents in governed enterprise knowledge for retrieval-augmented generation (RAG).
- Architected agentic-retrieval benchmarking across CPU, memory, throughput, and latency; proposed and shipped the **production billing model** for a new serverless enterprise search offering.
- Lead cross-team architecture reviews and technical decisions; mentor and onboard engineers across information retrieval, vector search, Azure OpenAI, and agent orchestration.

Software Engineer II

Jun. 2022 – Feb. 2025

Microsoft — Azure AI Search C++, C#, Java, Python

Redmond, WA

- Helped scale **vector search from 1 to N** and **quantization from 0 to 1, then 1 to N**, building and scaling capabilities across billions of embeddings.
- Drove **vector quantization from Public Preview through GA**, delivering **8–32x customer cost savings** and **up to 20x lower latency** through binary vectors, scalar and binary quantization, and SIMD-accelerated distance computation.
- Optimized vector distance kernels with **SIMD, loop unrolling, independent accumulators, and fused multiply-add**, improving instruction-level parallelism and throughput.
- Designed hybrid-search subscore fusion and score thresholding to improve relevance across blended vector and keyword retrieval.
- Designed a resource-aware quota mechanism for HNSW indexes that **reduced limit overshoot by 100x**.
- Owned an **extensible facet-aggregation engine** from specification through delivery, implementing a lexer, parser, and evaluator with BNF grammar, shunting-yard parsing, and Reverse Polish Notation.
- Root-caused cross-team production incidents, caught a critical quantization bug before release, and reviewed distributed-system and vector-algorithm changes for correctness and reliability.

Software Engineer

Jul. 2021 – May 2022

Microsoft — Azure AI Search

Redmond, WA

- Delivered index aliases to Public Preview, allowing applications to repoint to another search index with **no code changes or downtime**.
- Completed a phased, zero-downtime telemetry-table migration that preserved data integrity and produced a **50–100x query speedup**.

EARLIER MICROSOFT EXPERIENCE

- **Software Engineer Intern, Azure Cognitive Search** (Sep.–Dec. 2020): shipped the team's second most requested API and backend feature in C# to preview.
- **Software Engineer Intern, Azure Search** (Jun.–Aug. 2019): built a TypeScript search-site generator, query suggestions, filtering, and portal editing tools.
- **Software Developer Intern, Microsoft Garage** (Jan.–Apr. 2018): built an offline chest X-ray classification app, image pipeline, CI, iOS share extension, and TensorFlow Android integration.

SELECTED PROJECTS

- Human Pose Estimation Using Deep Neural Networks** [Code] *Jan. 2021 – May 2021*
- Led a student team building a heatmap-based 2D pose model on COCO 2017, **trained from scratch with random weight initialization**. Architected the cloud training pipeline, model, augmentation, visualization, and deployment.
- Semantic Road Segmentation Using Convolutional Neural Networks** [Code] *Jul. 2020 – Sep. 2020*
- Trained a U-Net CNN on KITTI Road using Keras, achieving **up to 99.1% F1** and 91% in the worst case.
 - Implemented the complete training pipeline, custom losses, data generation and augmentation, cloud training, evaluation, and architecture debugging.
- Self-Supervised Monocular Depth Estimation** [Code] *Jan. 2020 – May 2020*
- Adapted Monodepth2 to TensorFlow and **136,000 DrivingStereo image pairs** without ground-truth depth; used a ResNet-18 U-Net, stereo reconstruction, edge-aware smoothness, multi-scale loss, auto-masking, and minimum reprojection.
- Battlesnake Reinforcement Learning Agent** *Jan. 2019 – May 2019*
- Trained a keras-rl agent through self-play and competition against public agents to make real-time survival decisions in a multi-agent Battlesnake environment.

LEADERSHIP & COMMUNITY

- Industry Mentor, SENG 321 — Requirements Engineering** **2024; 2026 – Present**
University of Victoria *Victoria, BC*
- Mentor during Sep.–Dec. 2024 and since Aug. 2026. Authored challenges in LLM reliability and production AI; coach teams on requirements, scope, technical assumptions, and evaluation. One of 18 mentors for a 120-student cohort in 2024.
- Founder & Program Director, Seniors' Digital Literacy Program** **Jul. 2015 – Oct. 2021**
University of Victoria *Victoria, BC*
- Founded and directed the program for six years, growing it to **180 volunteers, 650+ seniors served, and 30 workshops**; secured funding, lab space, sponsorship, and administrative support.
 - Designed concurrent beginner and advanced curricula, built a recurring volunteer pipeline, and handed the institutionalized program to successors who sustained it beyond my tenure.
- Chair & Vice-Chair, IEEE Student Branch** **Sep. 2019 – Jan. 2021**
University of Victoria *Victoria, BC*
- Co-led a six-person executive and 70-member branch; created a **14-workshop series reaching 350+ students** across Git, machine learning, circuit design, and hardware prototyping.
 - Authored and taught curriculum, secured funding and lab access, recruited industry instructors, and established a recurring program that continued after my term.
- Founder & Conference Chair, Tech & Business Strategy Conference** **Aug. 2019 – Nov. 2019**
University of Victoria *Victoria, BC*
- Secured university support and led planning, logistics, speakers, and execution for a **200+ attendee conference** connecting technology and business strategy.

EDUCATION

- Bachelor of Engineering, Electrical & Computer Engineering** **Sep. 2016 – Apr. 2021**
University of Victoria Cumulative average: **97%** *Victoria, BC*
- Coursework and projects spanning distributed systems, algorithms, signal processing, embedded design, and machine learning.
 - Jamie Cassels Undergraduate Research Award recipient; researched hardware acceleration for machine-learning neural networks.

SELECTED AWARDS & CREDENTIALS

- Y Combinator AI Startup School — Inaugural Cohort** (Jun. 2025) — selected for an invitation-only gathering of approximately 2,500 AI researchers, engineers, and builders; **8% acceptance among 30,000 applicants**.
- Schulich Leader Scholarship** (2016) — \$80,000 full ride; **one of 50 awarded nationally, 3.3% acceptance among nominees** and 0.017% among all 300,000 high school graduates.
- National Champion, Michael Smith Science Challenge** (2014) — national-record 97.5% among 1,700+ candidates.
- Governor General's Academic Medal, Bronze** (2016).
- Jamie Cassels Undergraduate Research Award** (2019).
- 1st Place, Western Engineering Competition — Senior Design** (2020).
- 1st Place (3x), UVEC Engineering Competition — Senior Design** (2017–2019).